

Maximizing profits and expanding into new sources of revenue for a software company

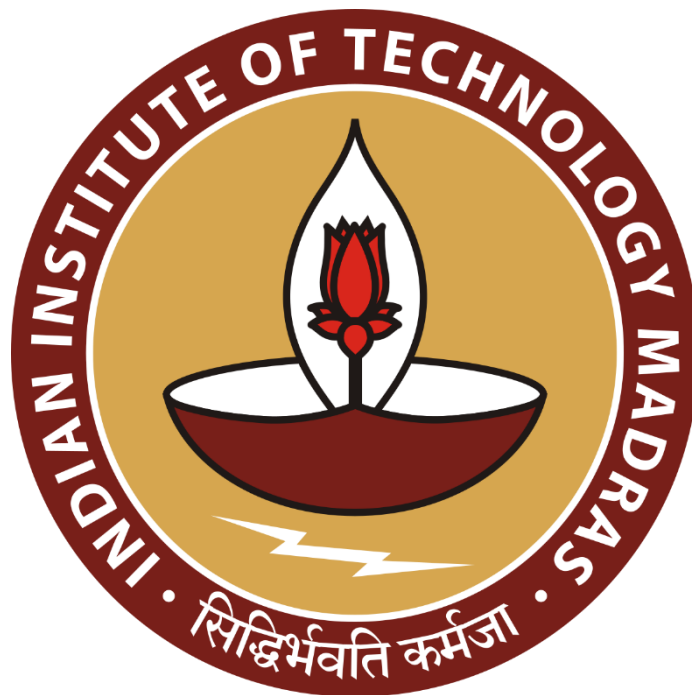
A Mid-Term report for the BDM capstone Project

Submitted by

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1. Executive Summary

BLISS Infosoftware Pvt Ltd is a software selling company in New Delhi, and faces some crucial problems in the running of their business such as dealing with their business partners who resell their products to life insurance agents across the country and not being able to expand into rural areas and D2C sector of revenue. So, in order to solve these problems, the company needs data driven analysis and solutions to maximize their profits and operational efficiency.

In this capstone project, the data generated by the business in the months of March, April, and May of 2025 will be collected and analyzed to provide insights and solutions for the issues the company faces. In this Mid-Term report, metadata and descriptive statistics regarding the important datasets needed for finding solutions of the company's problems will be listed. The proof of the company's existence and the originality of the data will also be provided in this report.

This data is meant to be cleaned and processed to obtain insights and the process for the same will be discussed in this report. The main tool for analysis is Microsoft Excel because of its easy to use nature and that the raw data provided from the Management Information System of the company generates reports in excel format.

The insights derived from this report will help the company in expanding into new areas revenue-wise and geographically and improve their crew-resource management.

2. Proof of originality

2.1 Interaction with owner

Here is the link for the video-

<https://youtu.be/XN9KU92AWhk>

2.2 Raw Data

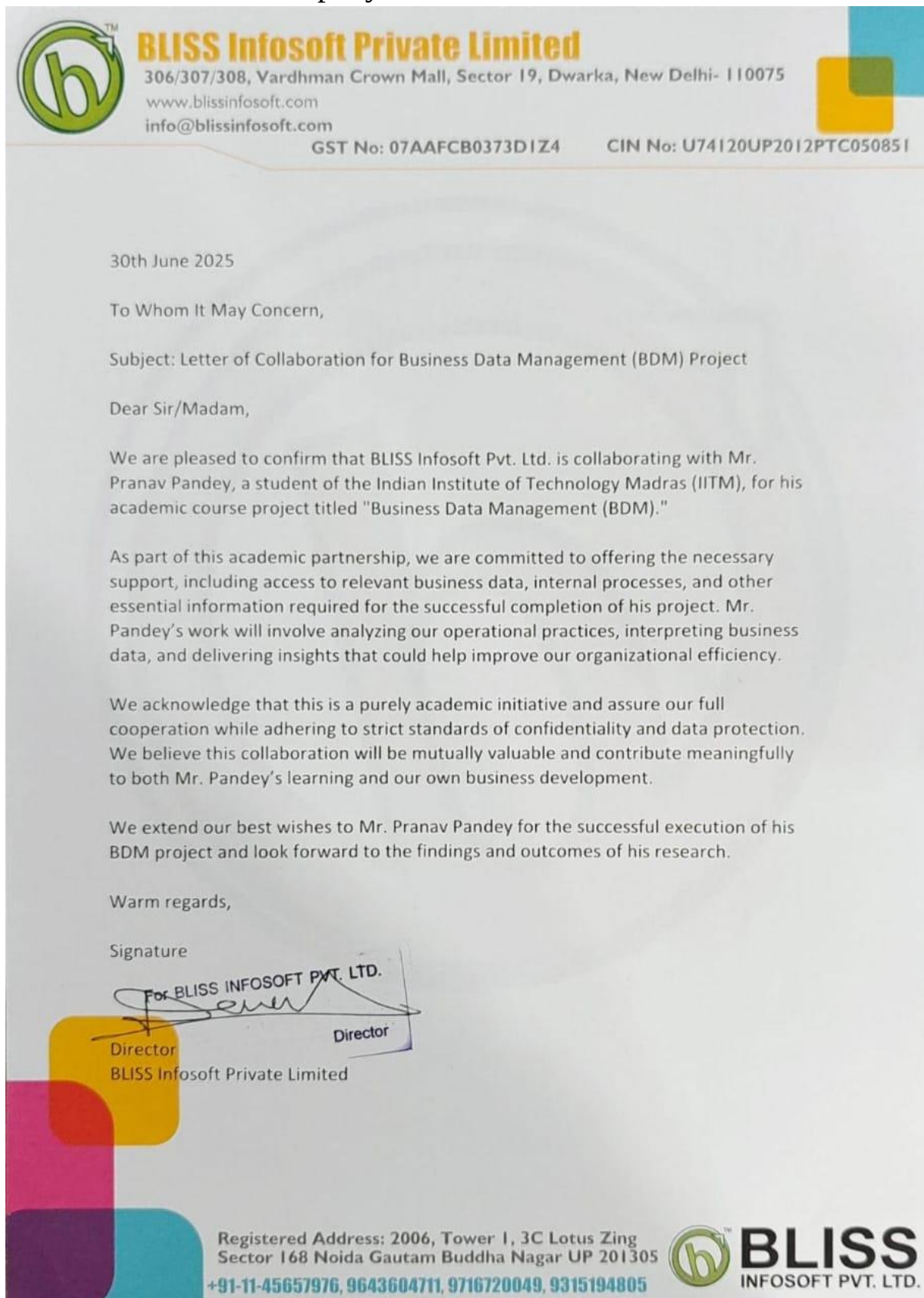
Here is the link for the raw data-

<https://drive.google.com/drive/folders/1kgXKiQEiu-xvmizNWCxoJxVHjXdR7oxT>

2.3 Images of the firm



2.4 Letter from the company



3. Metadata and Descriptive Statistics

3.1 Metadata

Link of Google Drive folder containing the raw data-

<https://drive.google.com/drive/folders/1kgXKiQEiu-xvmizNWCxoJxVHjXdR7oxT>

The raw data has been given to us by the Director of the company. The company uses a management information system also known as MIS to store all the data generated by the business. This system helps keep all the data organized and easy to access whenever in need.

Link of Google drive with cleaned data-

<https://drive.google.com/drive/folders/1g3LioqPPFefsA9DHI-A2f8DDMR0ioq2X>

Here are some key points about all the datasets-

- Format of the data- Microsoft Excel
- Range of the data provided- March 2025 to May 2025
- Currency used in the data- Indian Rupee [(INR) or (₹)]

Here is a list of the datasets provide and all the Metadata keys in them-

- Marketing Data- This dataset deals with the various sources of marketing that the company uses to spread awareness regarding its products and services towards potential future business partners (resellers) and life insurance agents.

Name of the key	Description
M_techniques	The different ways of marketing employed by the company
Cost	The amount of money spent on the technique in a calendar month

- List of employees- This dataset provides the name of the employees associated with the company and the department in which they work.

Name of the key	Description
Name	Name of the employee
Dept	Department in which the employee works

- List of Dealers- This dataset provides information regarding all the B2B partners and resellers and amount of revenue generated by each one of them.

Name of the key	Description
Customer Name	Name of the business partner
Group	Based on revenue the dealers are divided into groups
Division	The city/district where the dealer is located
Branch	It is a subsection of the division
Location	Description of the division i.e. urban or rural
Revenue	Amount of software sold by the particular dealer

3.2 Descriptive Statistics

A. Statistics of marketing expenditure of March, April and May of 2025

Marketing Technique	Mean	Std. Error	Median	Mode	Std. Dev.	Max	Min	Sum
Billboards	180,566.67	29,882.79	156,300.00	145,400.00	51,758.51	240,000.00	145,400.00	541,700.00
Community engagements	65,866.67	11,852.61	64,600.00	46,000.00	20,529.33	87,000.00	46,000.00	197,600.00
Digital Advertising	26,550.00	6,201.28	29,650.00	14,600.00	10,740.93	35,400.00	14,600.00	79,650.00
Newspapers	17,100.00	916.52	16,500.00	15,900.00	1,587.45	18,900.00	15,900.00	51,300.00
Pamphlets	5,620.00	2,207.35	4,760.00	2,300.00	3,823.24	9,800.00	2,300.00	16,860.00
Seminars	147,866.67	25,334.33	137,940.00	109,800.00	43,880.35	195,860.00	109,800.00	443,600.00
Total	443,570.00	48,874.17	404,010.00	385,940.00	84,652.54	540,760.00	385,940.00	1,330,710.00

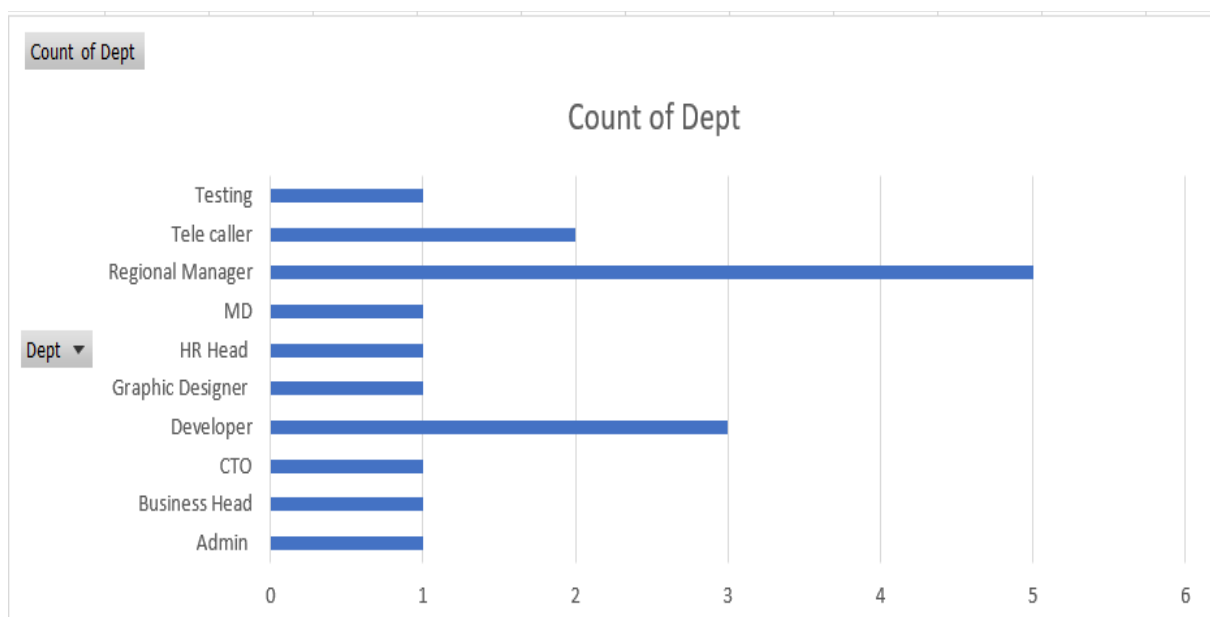
B. Statistics of revenue of all business partners month-wise

Month	Mean	Standard Error	Median	Mode	Standard Deviation	Maximum	Minimum	Count	Sum
March	135669.38	4921.0	137362.91	27824.59	56751.71	290694.39	27824.59	133	18044027.65
April	139554.14	5556.53	144230.92	25016.02	61373.83	291588.76	25016.02	122	17025604.82
May	139503.61	5803.13	145768.17	24522.05	61414.59	295210.43	24522.05	112	15624404.78

C. Split of urban and rural sales in the month of April

Location	Sum of Revenue
Rural	562344.42
Urban	16463260.4
Grand Total	17025604.82

D. Statistics of list of employees working for the company



4. Process of analysis

Here is a look into the steps followed to obtain meaningful insights from the raw data-

A. Collection of data-

After figuring out the problem statements, we need the relevant datasets to be analyzed to gain insights and solve problems. The following are the datasets collected-

- B2B partners list
- Marketing Data
- Employee structure

B. Cleaning the data obtained-

Once the data is obtained, it must be scanned to check for inconsistencies such as duplicates and invalid characters which may result in an error while making the charts for visualization. Also, any unwanted columns such as the contact info of the various business partners and salaries of the employees will have to be removed.

C. Understanding the data-

After cleaning the data, in this step we must understand the structure of the given data and its key features, these include-

- Descriptive statistics such as mean, median, mode, variance, etc.
- Simplifying the data using graphs, filters and other functions provided in Microsoft Excel to visualize it

D. Grouping and filtering the data-

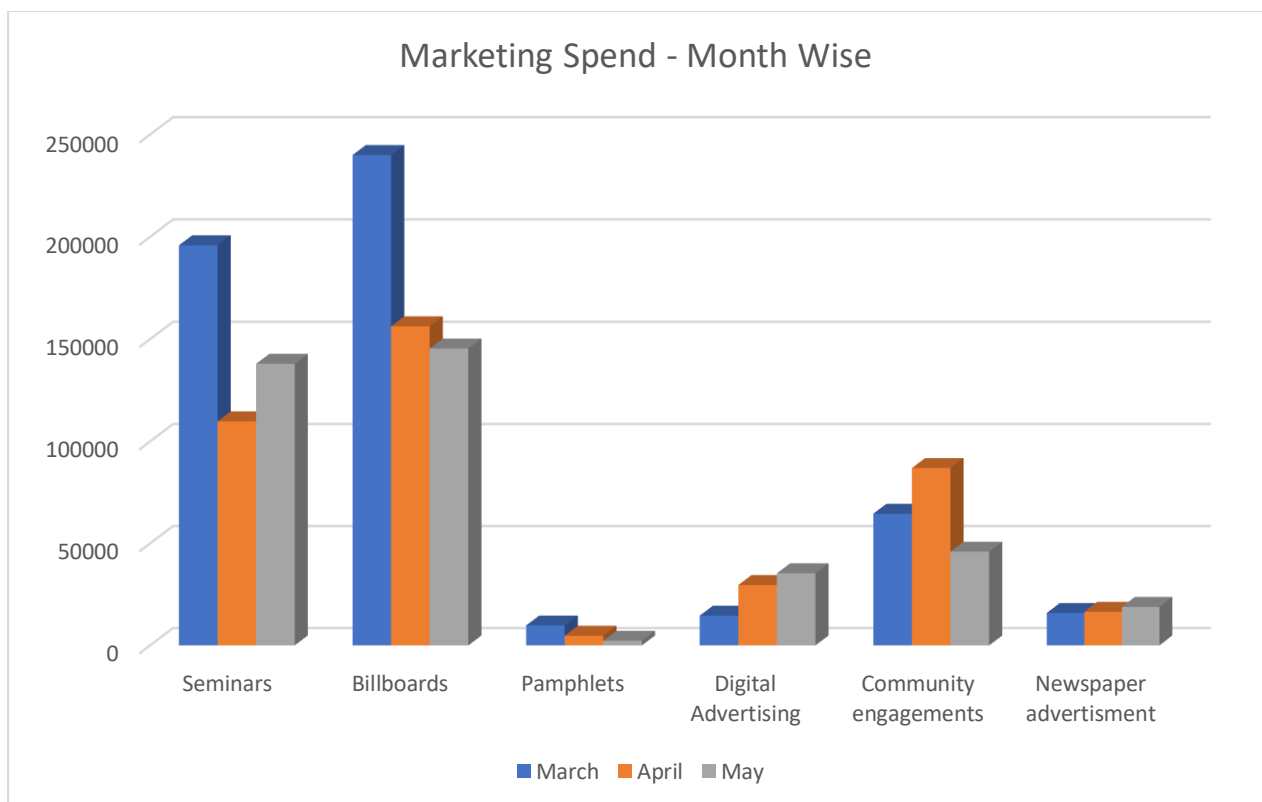
In this step, the different rows in the datasets will be grouped based on their characteristics. For example, the dealers will be grouped into urban and rural, the employees will be grouped based on their department, etc.

E. Interpreting the data and receiving insights-

In this step, we will visualize the data and try to identify its patterns and trends, in order to achieve valuable insights and derive concrete solutions.

5. Results and findings

A. Visualization of the marketing data-



About the graphs-

- The first graph tells us about the money spent on marketing by the company month on month.
- The second graph breaks down the sources of marketing used by the company and the amount spent on each one of them.

Key information obtained from the graphs-

- From the first graph, we can observe that the marketing expenditure of the company seems to be going down month on month as represented by the line.
- The second graph tells us that the company is increasing its focus mainly on digital advertising and newspaper advertisements while reducing spending on pamphlets and billboards.
- Also, the spending on seminars and community engagements seems to be fluctuating along the different months.

Analyzing the problem statement and linking it to the data-

- One of the problems was the inability of the company to expand into rural areas. Looking at the data, this makes sense as the means of marketing employed by the company are very urban-centric and not ideal for spreading the word across rural areas.
- Also, focusing marketing efforts solely on urban areas may eventually lead to market saturation, limiting the company's ability to generate significant sales growth.

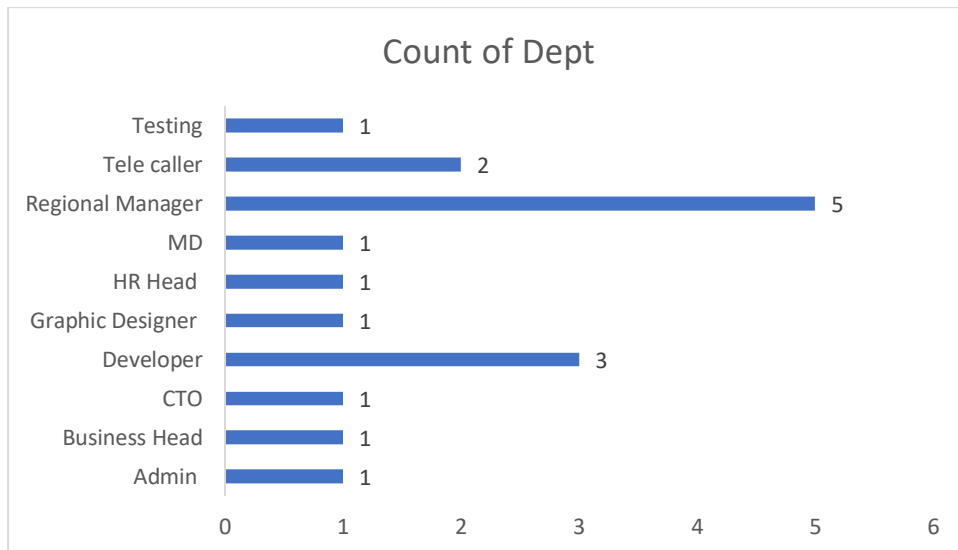
Possible solutions and insights to the problems-

- The increase of spending in the Digital advertisement sector may be counter-productive to the growth of the business in rural areas. Hence, an increase in its spending is okay but must be done so only after analyzing its key metrics such as, what type of people view these ads, their location, etc.
- The company should look into their existing customer base and try to find out the patterns in them such as, their background, where they live and how did they get to know about the company. This will help us further in trying to understand the productivity and effectiveness of each way of marketing and to spend on them accordingly.
- Newspaper advertisements are probably the best way to market a product as they are distributed quite evenly in urban and rural areas. And, since the job of a life insurance agent does not require any formal degree, it may be an opportunity to those living in rural area with a lack of education post the 12th standard.

Please note-

The findings of this report only analyzed with the help of the data regarding marketing and do not consider dealer/sales data. More insights and solutions will be presented in the final report where the sales data will also be thoroughly analyzed.

B. Visualization of the employee structure-



About the graph-

The above graph breaks down the different departments and roles taken up by the employees working for the company

Looking into the problem statement and link it with the data-

- The problem statement was regarding not having a dedicated support team to handle any issues that may arise after the app is deployed and a bug is caught by the users.
- In the organization structure, we can clearly see three developers and person in the testing department but there no support department.

Possible solutions and insights into the problem-

- One solution is to transfer the workload of a support team to the development team, whenever they are not involved in the making of a new product and having a dedicated number for technical support.
- The company should store some data regarding the number of hours an employee works and what they do in it. This will help the company in improving their crew-resource management.